# Time Series Tutorial 1

### Handing Time Series in R

```
sales <- read.csv("sales_data.csv")

# Run once - install anything missing
pkgs <- c("tidyverse", "tsibble", "feasts", "fable", "timetk",
          "modeltime", "kableExtra", "here")
```

### Handing Dates and Aggregation

```
# Load and filter data
df <- read.csv("sales_data.csv")
df$date <- as.Date(df$date)
df <- subset(df, store_id == "S001")

# Aggregate by Month using format()
agg_month <- aggregate(sales ~ format(date, "%Y-%m"), data = df, FUN = sum)

# Aggregate by Week (%U for week of the year)
agg_week <- aggregate(sales ~ format(date, "%Y-%U"), data = df, FUN = mean)

head(agg_month)
```

```
  format(date, "%Y-%m")   sales
1               2021-01 3160.42
2               2021-02 3122.53
3               2021-03 3899.81
4               2021-04 4186.90
```

```
5               2021-05 4687.30
6               2021-06 4777.31
```

```r
sales_ts <- ts(agg_month$sales, start = c(2021, 1), frequency = 12)

frequency(sales_ts)
```

```
[1] 12
```

```r
start(sales_ts)
```

```
[1] 2021    1
```

```r
end(sales_ts)
```

```
[1] 2023   12
```

```r
cycle(sales_ts)
```

```
     Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
2021   1   2   3   4   5   6   7   8   9  10  11  12
2022   1   2   3   4   5   6   7   8   9  10  11  12
2023   1   2   3   4   5   6   7   8   9  10  11  12
```

```r
time(sales_ts)
```

```
          Jan      Feb      Mar      Apr      May      Jun      Jul      Aug
2021 2021.000 2021.083 2021.167 2021.250 2021.333 2021.417 2021.500 2021.583
2022 2022.000 2022.083 2022.167 2022.250 2022.333 2022.417 2022.500 2022.583
2023 2023.000 2023.083 2023.167 2023.250 2023.333 2023.417 2023.500 2023.583
          Sep      Oct      Nov      Dec
2021 2021.667 2021.750 2021.833 2021.917
2022 2022.667 2022.750 2022.833 2022.917
2023 2023.667 2023.750 2023.833 2023.917
```

```r
window(sales_ts, start = c(2022, 1), end = c(2022, 12))
```

```
         Jan     Feb     Mar     Apr     May     Jun     Jul     Aug     Sep
2022 3813.14 3702.30 4403.61 4740.49 5231.29 5167.31 5428.39 5221.75 4747.05
         Oct     Nov     Dec
2022 4504.49 4373.27 4319.78
```

```
stats::lag(sales_ts, k = 1)
```

```
         Jan     Feb     Mar     Apr     May     Jun     Jul     Aug     Sep
2020
2021 3122.53 3899.81 4186.90 4687.30 4777.31 4718.29 4631.21 3992.56 3996.86
2022 3702.30 4403.61 4740.49 5231.29 5167.31 5428.39 5221.75 4747.05 4504.49
2023 4095.59 5027.84 5347.16 5661.57 5913.21 5996.13 5806.09 5034.01 5172.38
         Oct     Nov     Dec
2020                 3160.42
2021 3878.24 3763.64 3813.14
2022 4373.27 4319.78 4296.96
2023 4937.70 4957.90
```

```
diff(sales_ts)
```

```
         Jan     Feb     Mar     Apr     May     Jun     Jul     Aug     Sep
2021          -37.89  777.28  287.09  500.40   90.01  -59.02  -87.08 -638.65
2022   49.50 -110.84  701.31  336.88  490.80  -63.98  261.08 -206.64 -474.70
2023  -22.82 -201.37  932.25  319.32  314.41  251.64   82.92 -190.04 -772.08
         Oct     Nov     Dec
2021    4.30 -118.62 -114.60
2022 -242.56 -131.22  -53.49
2023  138.37 -234.68   20.20
```

```
diff(sales_ts, lag = 12)
```

```
        Jan    Feb    Mar    Apr    May    Jun    Jul    Aug    Sep    Oct
2022 652.72 579.77 503.80 553.59 543.99 390.00 710.10 590.54 754.49 507.63
2023 483.82 393.29 624.23 606.67 430.28 745.90 567.74 584.34 286.96 667.89
        Nov    Dec
2022 495.03 556.14
2023 564.43 638.12
```

```r
is.ts(sales_ts)
```

```
[1] TRUE
```

```r
is.na(sales_ts)
```

```
 [1] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
[13] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
[25] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
```

```r
na.omit(sales_ts)
```

```
         Jan     Feb     Mar     Apr     May     Jun     Jul     Aug     Sep
2021 3160.42 3122.53 3899.81 4186.90 4687.30 4777.31 4718.29 4631.21 3992.56
2022 3813.14 3702.30 4403.61 4740.49 5231.29 5167.31 5428.39 5221.75 4747.05
2023 4296.96 4095.59 5027.84 5347.16 5661.57 5913.21 5996.13 5806.09 5034.01
         Oct     Nov     Dec
2021 3996.86 3878.24 3763.64
2022 4504.49 4373.27 4319.78
2023 5172.38 4937.70 4957.90
```

### Helper conversions

```r
t <- time(sales_ts)

# Extract Year and Month
years <- floor(t)
months <- round(12 * (t - years)) + 1

data.frame(Decimal = as.numeric(t), Year = years, Month = months)
```

```
    Decimal Year Month
1  2021.000 2021     1
2  2021.083 2021     2
3  2021.167 2021     3
4  2021.250 2021     4
5  2021.333 2021     5
6  2021.417 2021     6
```

```
7  2021.500 2021     7
8  2021.583 2021     8
9  2021.667 2021     9
10 2021.750 2021    10
11 2021.833 2021    11
12 2021.917 2021    12
13 2022.000 2022     1
14 2022.083 2022     2
15 2022.167 2022     3
16 2022.250 2022     4
17 2022.333 2022     5
18 2022.417 2022     6
19 2022.500 2022     7
20 2022.583 2022     8
21 2022.667 2022     9
22 2022.750 2022    10
23 2022.833 2022    11
24 2022.917 2022    12
25 2023.000 2023     1
26 2023.083 2023     2
27 2023.167 2023     3
28 2023.250 2023     4
29 2023.333 2023     5
30 2023.417 2023     6
31 2023.500 2023     7
32 2023.583 2023     8
33 2023.667 2023     9
34 2023.750 2023    10
35 2023.833 2023    11
36 2023.917 2023    12
```

### Subsetting with window

```r
# Extract only the second half of 2023
sub_ts <- window(sales_ts, start = c(2023, 7), end = c(2023, 12))
print(sub_ts)
```

```
         Jul     Aug     Sep     Oct     Nov     Dec
2023 5996.13 5806.09 5034.01 5172.38 4937.70 4957.90
```

### Lags and differences

```
# Shift the time index forward by 1 period (does not move the values)
lagged_ts <- stats::lag(sales_ts, k = -1)

# First-order differencing (removes linear trend)
diff_ts <- diff(sales_ts, differences = 1)

# Seasonal differencing (removes a 12-month seasonal pattern)
seasonal_diff_ts <- diff(sales_ts, lag = 12)

diff_ts
```

```
         Jan     Feb     Mar     Apr     May     Jun     Jul     Aug     Sep
2021          -37.89  777.28  287.09  500.40   90.01  -59.02  -87.08 -638.65
2022   49.50 -110.84  701.31  336.88  490.80  -63.98  261.08 -206.64 -474.70
2023  -22.82 -201.37  932.25  319.32  314.41  251.64   82.92 -190.04 -772.08
         Oct     Nov     Dec
2021    4.30 -118.62 -114.60
2022 -242.56 -131.22  -53.49
2023  138.37 -234.68   20.20
```

### Combining Series: `ts.union()` and `ts.intersect()`

```
ts_a <- ts(1:10, start = c(2023, 1), frequency = 12)
ts_b <- ts(11:18, start = c(2023, 4), frequency = 12)

# Union: keeps full range, fills gaps with NA
combined_union <- ts.union(ts_a, ts_b)

# Intersect: keeps only the overlapping months
combined_intersect <- ts.intersect(ts_a, ts_b)

combined_intersect
```

```
         ts_a ts_b
Apr 2023    4   11
May 2023    5   12
Jun 2023    6   13
```

```
Jul 2023    7   14
Aug 2023    8   15
Sep 2023    9   16
Oct 2023   10   17
```

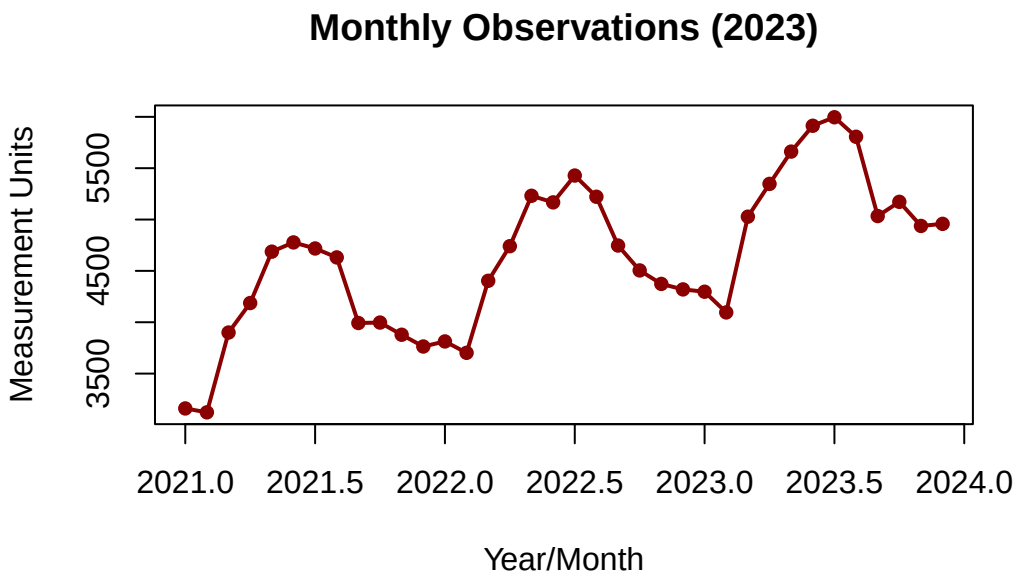
### Checking for missing values

```r
has_na <- anyNA(sales_ts)
cat("Contains missing values:", has_na, "\n")
```

```
Contains missing values: FALSE
```

### Time series plots

```r
plot.ts(sales_ts,
        main = "Monthly Observations (2023)",
        xlab = "Year/Month",
        ylab = "Measurement Units",
        col = "darkred",
        lwd = 2,
        lty = 1,
        type = "o",
        pch = 16)
```


Monthly Observations (2023)
Measurement Units
3500
4500
5500
2021.0
2021.5
2022.0
2022.5
2023.0
2023.5
2024.0
Year/Month